

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\tilde{\nu}_v \cdot \psi &= \left(\frac{g_u \times \psi}{|g_u \times \psi|} \right)_v \cdot \psi \\ &= \left(\frac{g_{uv} \times \psi + g_u \times \psi_v}{|g_u \times \psi|} \right) \cdot \psi + ((g_u \times \psi) \cdot \psi) \left(\frac{1}{|g_u \times \psi|} \right)_v \\ &= -\frac{\det(g_u, \psi, \psi_v)}{|g_u \times \psi|} \neq 0.\end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$L_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$< \chi, \chi' > = \prod_{r=1}^N \delta_{n_r n'_r} \cdot \prod_{r=1}^M \delta_{m_r m'_r} \cdot \prod_{r=1}^Q \delta_{q_r q'_r} \cdot \prod_{r=1}^T \delta_{t_r t'_r} \cdot \prod_{r=1}^4 \delta_{\nu_r \nu'_r}.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}g_v^+ &= v\psi, (\text{resp. } g_v^- = v\psi), \\ r_v &= v\psi_u, \\ \psi_v \cdot \psi &= \phi_v \cdot \phi, \\ \psi_v \cdot g_u^+ &= \phi_v \cdot f_u (\text{resp. } \psi_v \cdot g_u^- = \phi_v \cdot f_u), \\ \psi_v \cdot r_u &= (\phi \cdot f_{uu})_v - \frac{v}{2}(\phi \cdot \phi)_{uu} + v(\psi_u \cdot \psi_u),\end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$z \frac{d}{dz} + \begin{pmatrix} b_1(z) & b_2(z) & b_3(z) & \dots & b_{n-1}(z) & b_n(z) \\ 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \dots & 1 & 0 \end{pmatrix} \leftrightarrow \frac{d^n}{d\tau^n} + \bar{b}_1(\tau) \left(\frac{d^{n-1}}{d\tau^{n-1}} \right) + \dots + \bar{b}_n(\tau),$$